AR1 Simulation Comparison — Reference Panel at n=2000, m=4000

n=2000, m=4000, AR1 ρ=0.3, σ_s=0.3, 100 reps per F_{ST}

Scope of this report

- Companion to the main report, scaling the **reference-panel scenarios** (originally Sections 2–3, at n=500/m=1000) up to **n=2000, m=4000** — 4× the sample size and 4× the number of SNPs.
- Only the **bias-corrected** reference-panel configuration is run here (R's reference LD score matched to Python's correction, as in the original report's Figures 5–6) — the cleanest R-vs-Python comparison. The uncorrected variant was not scaled up.
- Each panel compares R's two reference-panel estimators (`res_reg_free_ref`, `res_reg_fixed_ref`) against three Python LDSC variants: `h2_ldsc_nw` (free intercept, unweighted), `h2_ldsc_fixed_nw` (fixed intercept, unweighted), and `h2_ldsc_free` — i.e. `h2_ldsc_reg`, the standard LDSC *weighted* free-intercept regression.
- **Implementation note:** R's reference-panel generator was rewritten to build each AR(1) column recursively ($X_{\cdot,1} \sim N(0,1)$, $X_{\cdot,j} = \rho X_{\cdot,j-1} + N(0, 1-\rho^2)$) instead of via an $m \times m$ eigen-decomposition and dense matrix square root. Both constructions give the identical stationary AR(1)(ρ) covariance (Var = 1, Corr$(i,j) = \rho^{|i-j|}$); the recursive form is $O(nm)$ instead of $O(nm^2 + m^3)$, which at $m = 4000$ is the difference between an estimated 5+ hours and about 15 minutes per scenario with R's reference BLAS. The per-replicate in-sample LD score (unused by this comparison) was also dropped from the reference-panel scripts to avoid its $O(nm^2)$ `crossprod(X)` cost on every one of the 1100 replicates.

Reference-panel comparison at n=2000, m=4000 (bias-corrected)

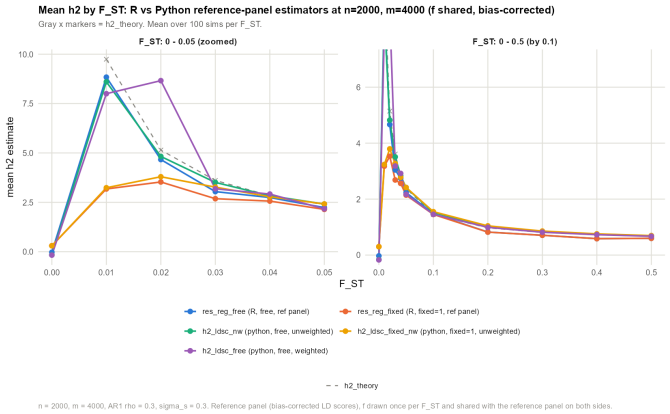


Figure 1. f shared between the reference panel and all 100 main replicates.

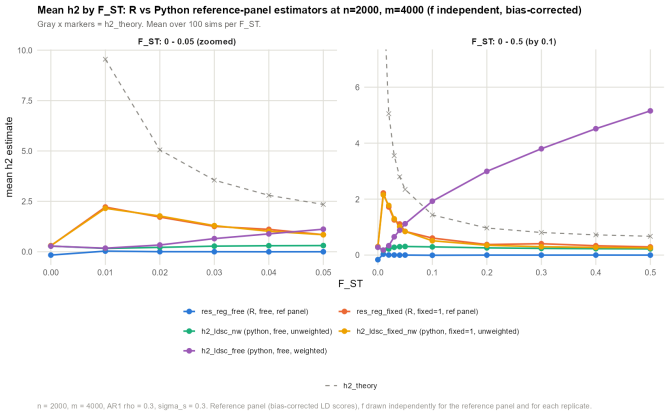


Figure 2. f drawn independently for the reference panel vs. each replicate.

See `report/report.tex` for the full n=500/m=1000 comparison (in-sample LD scores, the uncorrected reference-panel variant, and the diagnosis of what drove each gap) and `report/report_ldsc_free.tex` for the free-intercept-only n=500/m=1000 companion.